Jorge Alfredo Robles Calderón

Physicist — experimental materials, superconductivity and optics
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SUMMARY

Experimental physicist with independent experience in scientific software. Synthesis and characterization of spinel ferrites; a three-dimensional two-band superconductor solver designed, written and validated from scratch; early work in first-principles nonlinear optics and low-cost computational optics; web and data applications deployed and in use. B.Sc. Physics, Universidad Nacional de Colombia, degree conferred 5 November 2026.

Areas: condensed matter physics — superconductivity, functional materials, optics. Open to funded master's or PhD positions, and to research or industry roles in scientific software, data and materials R&D.

RESEARCH

Three-dimensional multiband TDGL solver — independent research, UNAL

Design, implementation and validation of a 3D two-band time-dependent Ginzburg–Landau solver for mesoscopic superconductors under field and transport current: gauge-invariant link variables, self-consistent vector potential, self-field current injection, Josephson voltmeter. Octave reference v13.3; Julia solver bit for bit against it; about 56 000 test assertions; lowest Landau level within 1.6 % and H_{c3}/H_{c2} within 0.05 %. Open source (MIT): **juliaTDGL**, two-band release 1.2 and N-band release 2.0 (Kramer–Watts-Tobin relaxation per band, finite interband quasiparticle exchange, anisotropic bands, thermal noise), github.com/georgeotjg/juliaTDGL.

Silent failure modes in three-dimensional two-band time-dependent Ginzburg–Landau simulations of geometric superconducting diodes — J. A. Robles Calderón, preprint, Zenodo (2026), doi:10.5281/zenodo.23163207. Eighteen failure modes, each yielding a converged but wrong quantity, each with a test; symmetry-exact diode control; diode efficiency bracketed from −47 % to −21 % [−33, −7] across padding, no converged value claimed. Second paper, manuscript under review: *Rectification in a notched two-band box at low field: two screening regimes and the terms that leave it unchanged*. Two further manuscripts in preparation.

$Zn_{1-x}Mg_xFe_2O_4$ spinel ferrites — bachelor's thesis (trabajo de grado), UNAL, 2026 (152 pp)

Advisor: D. A. Landínez Téllez. Synthesis of the series by glycine-assisted combustion; XRD with Rietveld refinement, SEM/EDX, UV-Vis diffuse reflectance. Analysis automated in Python, including band-gap fits whose window is selected by robustness (scikit-learn). Single-phase $Fd\bar{3}m$ solid solution, coherent domains of 20–26 nm, direct optical gap 1.88–1.91 eV nearly independent of composition. GGA+U band structures (treatment proposed and analysed in the thesis, computed within the research group) place the gap in the Fe–O subsystem.

Nonlinear optics of Weyl semimetals — started 2026

First-principles photocurrents in Weyl semimetals: Octopus (real-time TDDFT) and Quantum ESPRESSO + Wannier90 + Julia. Real-time lattice model reproducing the quantized circular photogalvanic response; Octopus validated on silicon including a symmetry null test; GaAs Wannier chain running.

Computational lensmeter — design stage

Measurement of spectacle lenses (sphere, cylinder, axis) by transmission deflectometry with a phone camera's RAW sensor and a laptop screen; elliptic inverse solver and thick-lens model.

APPLIED PROJECTS

Geospatial field-data platform. GPS-tagged forms, maps and spatial queries; Next.js, Supabase (PostgreSQL/PostGIS), Leaflet, PWA with offline queue. Deployed and used in the field for interviews, surveys and territorial records.

OCR + local language model. National Astronomical Observatory, UNAL: metadata from scanned documents, with validation that rejects malformed extractions.

Shape editor and run manager for the superconductor solver (JavaScript, Python): 3D sample shapes with live symmetry checks; terminal menu to launch, follow and resume long runs.

Commercial catalogue site. Next.js, TypeScript and Tailwind on Vercel, quotations over WhatsApp. tecnimotos-villamil.vercel.app

EXPERIENCE

Administrative and Technical Assistant — National Astronomical Observatory, Universidad Nacional de Colombia · 2023–2026
Extension and Technology-Transfer Assistant — Department of Physics, Universidad Nacional de Colombia · 2024–2025
Academic and Scientific Support — Universidad Nacional de Colombia · 2022–2025
Customer Operations — Iké Asistencia · 2023

EDUCATION

B.Sc. Physics — Universidad Nacional de Colombia, Bogotá · 2019–2026. Degree conferred 5 November 2026. Bachelor's thesis graded 5.0/5.0; GPA 4.0/5.0.
Technical degree, Administrative Assistance — SENA · 2019

SKILLS

Computation: Python (NumPy, SciPy, pandas, scikit-learn, PyTorch, Matplotlib), Julia, MATLAB/Octave, SQL, TypeScript, LaTeX, Git, Linux.
Numerical methods: finite differences on staggered grids, gauge-invariant discretisation, IMEX schemes, Poisson/elliptic solves, preconditioned Krylov solvers, regression and acceptance testing.
First principles: DFT/DFT+U concepts and band-structure analysis; Quantum ESPRESSO and Octopus (real-time TDDFT), in current use.
Laboratory: combustion synthesis, XRD with Rietveld refinement, SEM/EDX, UV-Vis diffuse reflectance.
Data and cloud: data analysis and fitting pipelines in Python; SQL (PostgreSQL), AWS, Docker.
GIS: PostGIS, spatial queries, Leaflet, GPS field data.
Web: Next.js/TypeScript, Supabase (deployed applications).
Languages: Spanish (native), English (B2).